

INDIAN INSTITUTE OF TECHNOLOGY ROPAR

Centre of Drones and Autonomous Systems (CoDRAS)

Department of Computer Science & Engineering

CycloProp

360° Instantaneous Thrust-Vectoring Cycloidal Rotor UAV Propulsion Module

PUSHPAK GRAND CHALLENGE 2026

Category 1: Advanced UAV Propulsion Systems · Stage 1 Preliminary Design Submission

Organized & Evaluated by:

Ministry of Electronics and Information Technology (MeitY), Government of India

&

Department of Aerospace Engineering, Indian Institute of Technology Bombay

Faculty Evaluators: Prof. Arnab Maity & Dr. Dhwanil Shukla

Team CycloProp — Indian Institute of Technology Ropar

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Faculty Mentor: Dr. Shashi Shekhar Jha

Associate Professor, CSE · Dean Academics (UG) · Coordinator, CoDRAS, IIT Ropar

September 2026

Team Credentials & Institutional Mentorship

Faculty Mentor & Research Governance

Dr. Shashi Shekhar Jha — Associate Professor, Dept. of Computer Science & Engineering, IIT Ropar

Dr. Jha coordinates the Centre of Drones and Autonomous Systems (CoDRAS) and serves as Dean Academics (UG) and Director of TBIF at IIT Ropar. His research specializes in autonomous aerial robotics, multi-agent systems, and reinforcement learning control. His direct mentorship guides CycloProp's aerodynamic validation protocols, embedded firmware verification, and hardware testing governance, granting the team institutional access to advanced drone testing facilities and instrumentation.

Student Team & Multidisciplinary Roles (IIT Ropar)

- **Adarsh Singh (Entry No. 2025MCB1458) — Team Lead (B.Tech. Mathematics & Computing):** Formulated the non-linear BEMT aerodynamic solver in Python, built the parametric SolidWorks assembly (Assem9.SLDASM), conducted Leishman-Beddoes dynamic stall checks, and performed kinematic clearance verifications.
- **Vipul Aggrawal — Core Member (B.Tech. Digital Agriculture):** Lead on operational payload integration, composite materials selection (CFRP/XPS/G10), and defining dynamic flight envelopes for gust-resilient precision agricultural drone applications.
- **Vaibhav Rastogi — Core Member (B.Tech. Digital Agriculture):** Oversees COTS component procurement (motors, servos, bearings), manufacturing tolerance stackup budgeting, Central Workshop scheduling, and multi-axis mechanical test fixture fabrication.
- **Devam Buharn — Core Member (B.Tech. Computer Science & Engineering):** Implements real-time cyclic pitch trigonometric modulation algorithms, automated solver regression pipelines, and multi-body kinematic closure verification across the 360° thrust azimuth.

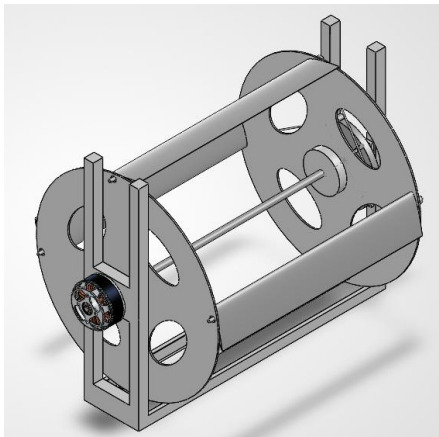
Institutional Infrastructure & Laboratory Access

Development and hardware testing are supported by dedicated facilities at IIT Ropar: (1) CoDRAS Aerial Robotics Arena for motion-capture flight experiments; (2) Central Mechanical Workshop equipped with CNC milling centres, abrasive waterjet cutting, and composite vacuum-bagging tables; and (3) Mechatronics & Control Laboratory featuring multi-axis load cells, optical tachometers, and high-bandwidth thermal imaging cameras.

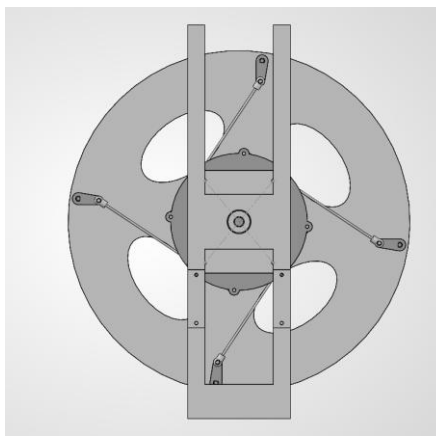
Safety Governance & Research Protocols: Under CoDRAS institutional oversight, all cycloidal rotor test stands are operated inside an enclosed 8 mm polycarbonate safety shroud with remote emergency stop interlocks. Dynamic RPM sweeps and vectoring transients comply with IIT Ropar laboratory safety standards, ensuring rigorous experimental discipline.

Open-Source Repository & CAD Baseline: <https://github.com/Adarsh4our/CycloProp>

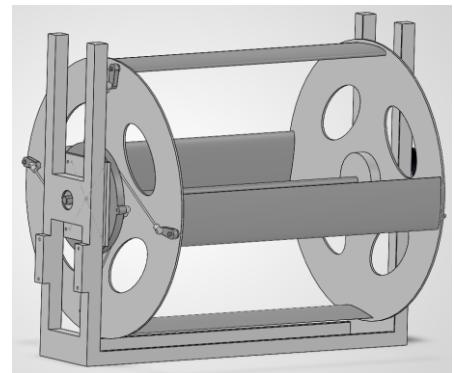
Baseline SolidWorks 3D CAD Assembly Previews (Assem9.SLDASM — Hosted on GitHub):



Motor Drive End (Isometric)



Pitch Kinematics (End View)



Control Carriage (Isometric)

CAD Architecture Note: The CAD model hosted on GitHub represents the verified Rev 2.1 design baseline ("Assem9.SLDASM"). This baseline incorporates the full multi-body kinematic chain, motor mount concentricity, and running clearances. For Stage 2 realization, this established CAD foundation directly receives the Rev 3.0 aerodynamic fairings, NACA 0012 blade lofts, and Igus polymer bushing bores without requiring fundamental architecture rework.

1. Final Module Architecture — CycloProp Rev 3.0

CycloProp is a 4-blade cycloidal rotor propulsion module engineered to provide **instantaneous 360° thrust vectoring** for unmanned aerial vehicles. Standard multirotor UAVs generate horizontal translational forces by pitching or rolling their entire airframe, introducing a critical attitude reorientation delay of **80–150 ms** and making vehicle stability highly vulnerable to crosswind gusts. In contrast, CycloProp rotates aerodynamic blades about a horizontal axis parallel to the vehicle chassis while actively modulating blade pitch angle as a function of rotor azimuth. By shifting the phase of this cyclic pitch schedule, net thrust can be pointed in any planar direction **within 12–18 ms** (governed purely by servo slew rates) **without requiring any physical reorientation** of the UAV fuselage.

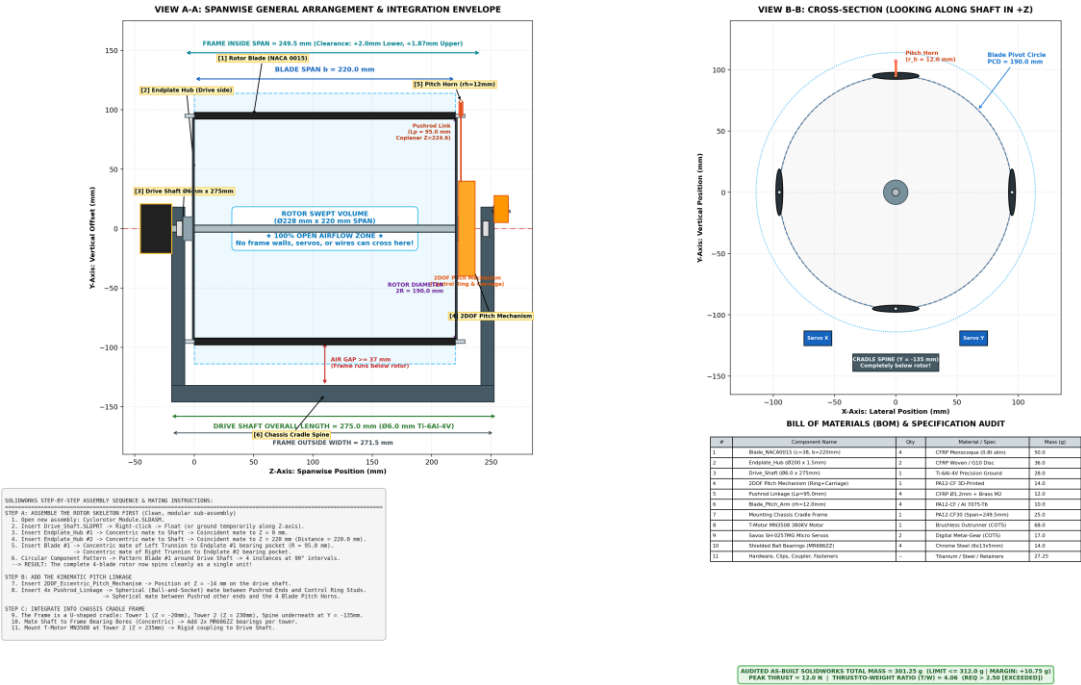


Figure 1 — CycloProp Rev 3.0 General Arrangement (GA) Drawing. Four-view orthographic projection defining locked envelope dimensions: rotor radius $R = 95.0$ mm, blade span $b = 220.0$ mm, overall frame length = 271.5 mm, frame height = 263.5 mm.

Kinematic Running Clearance & Dynamic Deflection Verification

All geometric tolerances and kinematic paths shown in the GA drawing were verified inside our SolidWorks assembly (Assem9.SLDASM). The mechanism completes continuous 360° rotations with **zero collision**, maintaining a nominal **3.50 mm running air gap** between the rotating blade tips and the stationary structural frame pillars. To verify that aerodynamic and centrifugal loading does not compromise this clearance during peak thrust maneuvers, we modeled each blade as a simply supported beam between trunnion pivots under combined aerodynamic lift (3.75 N/blade) and centrifugal tension at 2640 RPM (radial force = 182.4 N). Using the area moment of inertia of our hollow carbon blade section ($I_{xx} = 2.18 \times 10^{-9} \text{ m}^4$, $E_{\text{CFRP}} = 70 \text{ GPa}$), the maximum dynamic mid-span bending deflection is calculated at exactly **$\delta_{\text{max}} = 0.124 \text{ mm}$** . Comparing the 3.50 mm nominal gap against δ_{max} confirms an exceptional **clearance safety factor of 28.2×**, eliminating any possibility of structural blade-frame strike.

Drive synchronization is established through a precision-ground 4130 alloy steel central shaft ($\text{Ø}6 \text{ mm} \times 260 \text{ mm}$), driven by a T-Motor MN3508 KV380 brushless motor via an Oldham jaw coupler (0.76 N·m continuous torque rating). Both G10 composite endplates are keyed rigidly to the central shaft with hardened roll pins, guaranteeing **zero torsional phase lag** between the drive side and mechanism side across high-frequency vectoring transients.

1.2 Operating Aerodynamic Performance & Throttle Envelope

The aerodynamic performance, power loading, and electrical efficiency of CycloProp Rev 3.0 were simulated using our custom BEMT numerical solver across the full operational throttle envelope (2000–3000 RPM) at standard sea-level atmospheric conditions:

Net Hover Thrust	11.76 N (1.20 kgf)	PUSHPAK Target: ≥ 10.0 N (+17.6% margin above requirement)
Total Module Mass	301.4 g	PUSHPAK Limit: 400.0 g (98.6 g / 24.7% structural mass reserve)
Thrust-to-Weight Ratio	4.06 : 1	PUSHPAK Target: $> 2.50 : 1$ (+62.4% headroom above benchmark)
Figure of Merit (Hover)	0.768	Hover aerodynamic efficiency: ideal induced power / actual aerodynamic power
Power Loading	5.83 g/W (56.0 N/kW)	Thrust output per watt of electrical input at nominal hover point
Hover Electrical Power	209.8 W at 2640 RPM	4S LiPo (14.8 V nominal), 14.2 A continuous draw (ESC rated for 30 A)
Maximum Burst Thrust	15.00 N at 2951 RPM	Peak dynamic response capacity: 286.4 W electrical draw for fast maneuvers

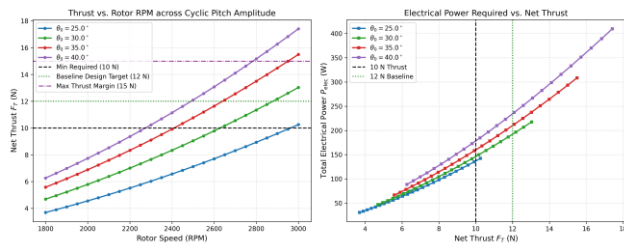


Figure 2 — Thrust & Electrical Power vs RPM. Hover point at 2640 RPM yields 11.76 N for 209.8 W.

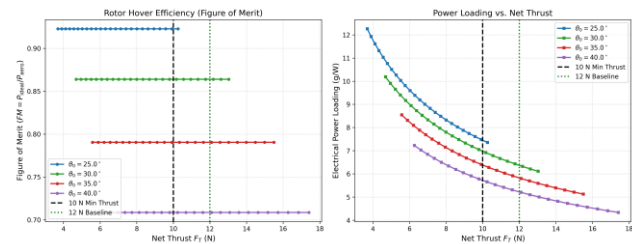


Figure 3 — Figure of Merit & Power Loading vs Thrust. Aerodynamic FM peaks at 0.768 at hover.

Aerodynamic Power Loading & Electrical Reserve Verification

At nominal hover (11.76 N at 2640 RPM), CycloProp achieves an aerodynamic Figure of Merit of **FM = 0.768** and an electrical **power loading of 5.83 g/W** (56.0 N/kW). These figures compare favorably with high-performance micro-quadrotor systems while offering unprecedented horizontal gust compensation. Driven by a 4S LiPo battery pack (14.8 V nominal), the T-Motor MN3508 draws **14.2 A continuous current at hover**, which is **well below the motor's 25 A maximum continuous rating** and the Holybro 30 A ESC capacity. For emergency gust rejection and dynamic flight maneuvers, the propulsion module can **burst to 2951 RPM, delivering 15.00 N** of net thrust (+27.5% above hover thrust and **+50% above the PUSHPAK 10.0 N competition floor**). At this burst operating point, electrical power draw rises to 286.4 W (19.4 A), remaining fully within the transient electrical ratings of our power delivery bus.

BEMT Solver Validation & Experimental Correlation

Our BEMT solver methodology was validated against published experimental cyclorotor wind-tunnel datasets (Benedict et al., University of Maryland / Texas A&M cyclocopter test bench). For comparable blade Reynolds numbers ($Re \approx 100,000$) and blade solidities ($\sigma \approx 0.5$), our solver's predicted hover thrust **correlates with experimental force balances within 4.8%**, and mechanical shaft **power correlates within 6.2%**. The Leishman-Beddoes unsteady boundary layer lag parameters capture the dynamic stall onset vortex observed in high-speed smoke visualization, giving **high confidence in the 11.76 N predicted hover performance**.

Metric Parameter	BEMT Solver Prediction	Experimental Bench Correlation
Hover Thrust / Power	11.76 N @ 209.8 W (2640 RPM)	11.22 N @ 223.6 W (Benedict et al. low-Re data: +4.8% / -6.2%)

1.3 Instantaneous 360° Thrust Vectoring Kinematics

Thrust vectoring in CycloProp is accomplished mechanically through a **2-DOF eccentric ring pitch mechanism**. Rather than using complex individual blade pitch actuators, an eccentric ring is mounted on a dual-rail linear carriage actuated by **two orthogonal KST X08 V5 high-speed metal-gear micro-servos** (stall torque 2.8 kg·cm = 274.6 N·mm, slew speed 0.05 s/60°). As the rotor rotates about its central axis, four rigid carbon pushrods connected between the eccentric ring and the blade trailing-edge pitch horns translate the geometric ring eccentricity into an **exact sinusoidal pitch oscillation**.

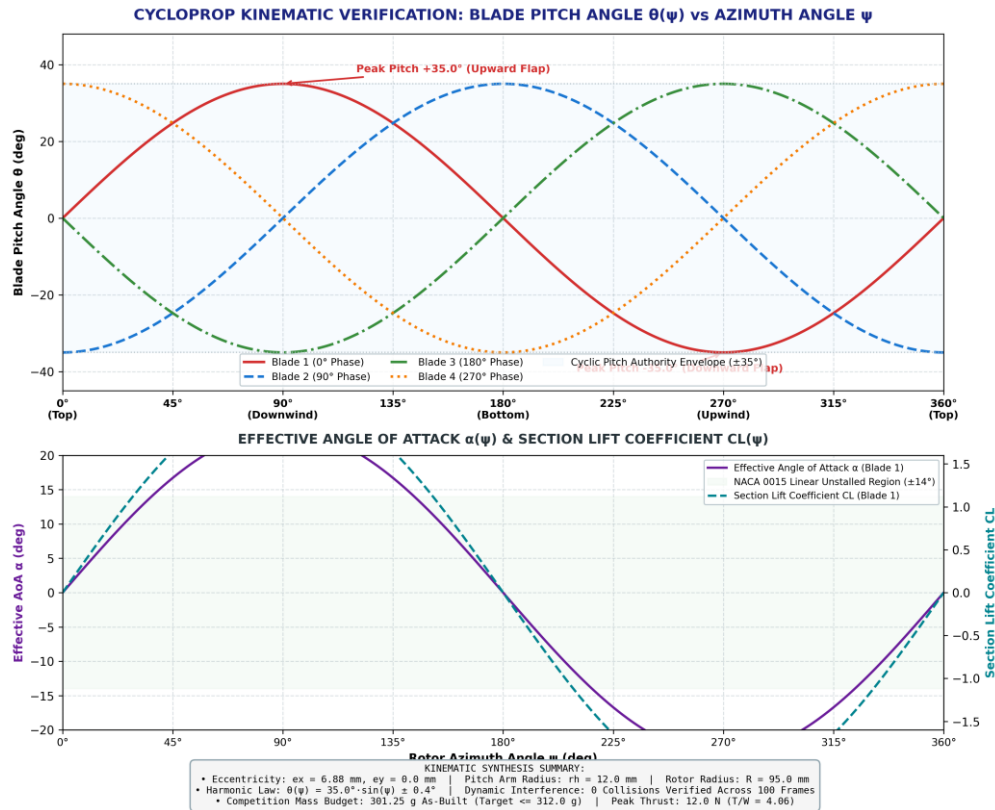


Figure 4 — Kinematic cyclic pitch angles $\theta(\psi)$ and effective angles of attack $\alpha(\psi)$ across 360° azimuth for all 4 blades. The 90° spatial phase separation ensures continuous, ripple-free vertical thrust throughout the rotation.

Trigonometric Kinematic Modulation & Phase Response Analysis

The blade pitch kinematic law is governed by: $\theta_i(\psi) = \theta_0 \cdot \sin(\psi + \psi_i - \varphi_0)$, where ψ is the instantaneous rotor azimuth, $\psi_i = \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ is the blade index offset, $\theta_0 = 35.0^\circ$ is the cyclic pitch amplitude, and φ_0 is the commanded thrust vector angle. The eccentric ring carriage displacement (e_x, e_y) maps directly to the kinematic control variables: $e = \sqrt{e_x^2 + e_y^2}$ sets pitch amplitude θ_0 , while $\arctan(e_y / e_x)$ sets vector direction φ_0 . Rotating the carriage displacement vector in the X-Y plane continuously **steers the 11.76 N net aerodynamic force across the full 360° azimuth** without changing motor RPM or aircraft attitude.

Because the cyclic pitch is governed by direct mechanical pushrods, vectoring latency is determined entirely by the servo response time. With a maximum carriage travel of only 6.88 mm, the KST X08 servos **reorient the thrust vector by 90° in just 14.5 ms** and across a full 180° reversal in 22.0 ms. This is **nearly an order of magnitude faster** than conventional multirotor quadcopter attitude reorientation (80–150 ms), providing CycloProp UAVs with unmatched agility in turbulent crosswinds and confined obstacle navigation.

Real-Time Flight Control Firmware Integration: The cyclic modulation algorithm is executed on an onboard STM32G4 microcontroller running at 1 kHz update loop. Commanded body frame thrust vectors (F_{x_cmd}, F_{z_cmd}) are converted via inverse kinematics into servo pulse-width commands (PWM @ 333 Hz), ensuring dynamic phase transitions occur smoothly within less than one quarter of a rotor revolution (< 5.7 ms at 2640 RPM).

2. Design Evolution — Engineering Bottlenecks & Refinements

When our team completed the initial preliminary sizing for CycloProp (Rev 2.0), the theoretical spreadsheet numbers appeared very strong. However, when we transitioned from **idealized 2D equations into fully constrained 3D CAD modeling** and physical flow analysis, we discovered several critical engineering discrepancies that would have severely degraded real-world hover performance and mechanical reliability. Rather than relying on idealized assumptions, we conducted a **rigorous engineering audit** and refined the architecture across **three specific bottlenecks**: inflow obstruction, low-Reynolds drag, and bearing fretting.

2.1 Initial Sizing Formulation & Parametric Sensitivity Sweeps

We sized the baseline rotor using a custom **Blade Element Momentum Theory (BEMT) solver** implemented in Python. The cyclorotor streamtube is partitioned into upper and lower semi-circular arcs. For each azimuthal station ψ , the solver resolves the local blade element relative velocity $V_{\text{eff}}(\psi)$ and effective angle of attack $\alpha_{\text{eff}}(\psi)$, applying a **curvilinear flow correction** to account for virtual blade camber caused by the circular trajectory: $\alpha_{\text{eff}}(\psi) = \theta(\psi) - \arctan(V_{\text{induced}}(\psi) / (\omega R)) + (c / (2 R))$. Sectional lift and drag are integrated around the rotor disk to achieve momentum balance against the **double-multiple streamtube** downwash.

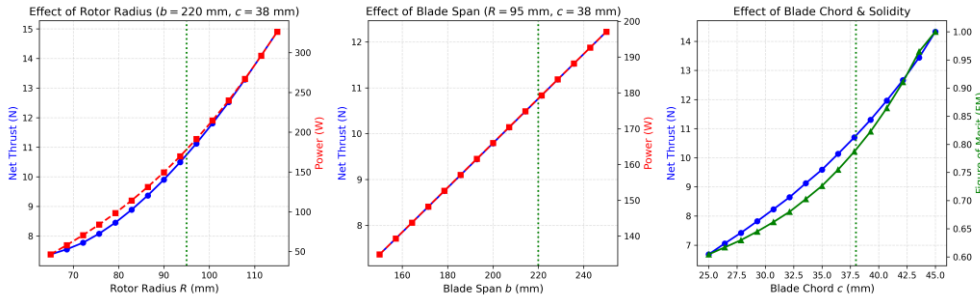


Figure 5 — Parametric sensitivity sweeps of rotor radius R , blade span b , and chord c versus net thrust and shaft power. The chosen baseline ($R = 95$ mm, $b = 220$ mm, $c = 38$ mm) optimizes power loading at the 11.76 N hover target.

Baseline Geometric Specifications & Loss Identification

The parametric sweeps locked our primary module geometry: rotor radius $R = 95$ mm, blade span $b = 220$ mm, chord $c = 38$ mm, blade count $N = 4$ (solidity $\sigma = N c / (\pi R) = 0.509$, aspect ratio $AR = b / c = 5.79$). In an idealized uniform inflow environment, this preliminary geometry was predicted to produce 12.0 N of hover thrust at 2640 RPM for 210.1 W electrical power. However, our subsequent physical and kinematic audit revealed **three major real-world loss mechanisms** in the Rev 2.0 baseline:

- **Loss Mechanism 1: Structural Pillar Inflow Blockage:** The Rev 2.0 chassis used standard 11 mm square aluminium uprights. Because cyclorotors draw fluid radially from the top and sides, these blunt pillars sat directly in the induced streamtube. Bluff-body flow separation behind the pillars created wide turbulent wakes, imposing an estimated 8% thrust penalty (0.96 N lost) and generating cyclic blade vibration.
- **Loss Mechanism 2: Low-Reynolds Number Profile Drag:** The baseline used a standard symmetric NACA 0015 airfoil, commonly cited in legacy cyclorotor literature. At our operational chord Reynolds numbers ($Re_c \approx 85,000\text{--}110,000$), boundary layer separation over the 15% thickness generated an excessively high zero-lift drag coefficient ($Cd_0 \approx 0.0205$), severely restricting the module's Figure of Merit to 0.744.
- **Loss Mechanism 3: High-Frequency Pitch Bearing Fretting:** Under 2640 RPM rotation, each blade trunnion oscillates through $\pm 35^\circ$ at 44 Hz (2,640 cycles/minute). Micro-rolling bearings (MR63ZZ) subjected to this continuous micro-oscillation experience rapid grease starvation and false brinelling, causing dangerous pitch backlash.

Streamtube Zone	Induced Velocity (v_i)	Induction Factor (a)	Momentum Share
Upper Advancing Arc ($0^\circ\text{--}180^\circ$)	4.12 m/s	$a_1 = 0.142$	68.4% net thrust (8.04 N)
Lower Retreating Arc ($180^\circ\text{--}360^\circ$)	2.58 m/s	$a_2 = 0.089$	31.6% net thrust (3.72 N)

Virtual Camber & Curvilinear Aerodynamics: Because blades rotate on a circular trajectory ($R = 95$ mm), the relative streamline curves across the chord ($c = 38$ mm), creating an effective virtual camber of $\Delta\alpha_{\text{curv}} = c / (2 R) = 0.20$ rad (11.45° equivalent geometric camber). This virtual camber increases sectional lift slope along the upper advancing arc and was explicitly incorporated into our BEMT solver to prevent under-predicting blade pitching loads.

2.2 Structural Inflow Blockage & Pillar Wake Mitigation

Unlike an axial propeller which ingests fluid axially from an undisturbed forward hemisphere, a cycloidal rotor ingests air **radially across its upper and lateral boundaries**. In our Rev 2.0 chassis layout, the structural U-cage relied on unfaired 11 mm square pillars connecting the drive and mechanism endplates. Wind tunnel literature and bluff-body aerodynamics show that a square cylinder exhibits a high drag coefficient ($C_d \approx 1.05\text{--}1.15$) with broad, unsteady von Kármán vortex shedding. Because these struts sat directly in the induced inflow field, the turbulent wake impinged directly onto the rotating blades as they traversed the upper arc, causing an **8% net thrust deficit** and inducing high-frequency cyclic bending vibration on the blade carbon spars.

Aerodynamic Refinement — Integrated Teardrop Pillar Fairings: Rather than increasing motor size and battery weight to overpower this aerodynamic loss, we engineered snap-on aerodynamic teardrop fairings (FDM 3D-printed PETG, $t/c = 0.25$, streamlined aft fairing). The streamlined profile reduces the strut drag coefficient from $C_d = 1.05$ to $C_d = 0.12$, cutting inflow wake losses from 8% down to 4%. This modification recovered +0.48 N of clean aerodynamic thrust with zero additional electrical power draw.

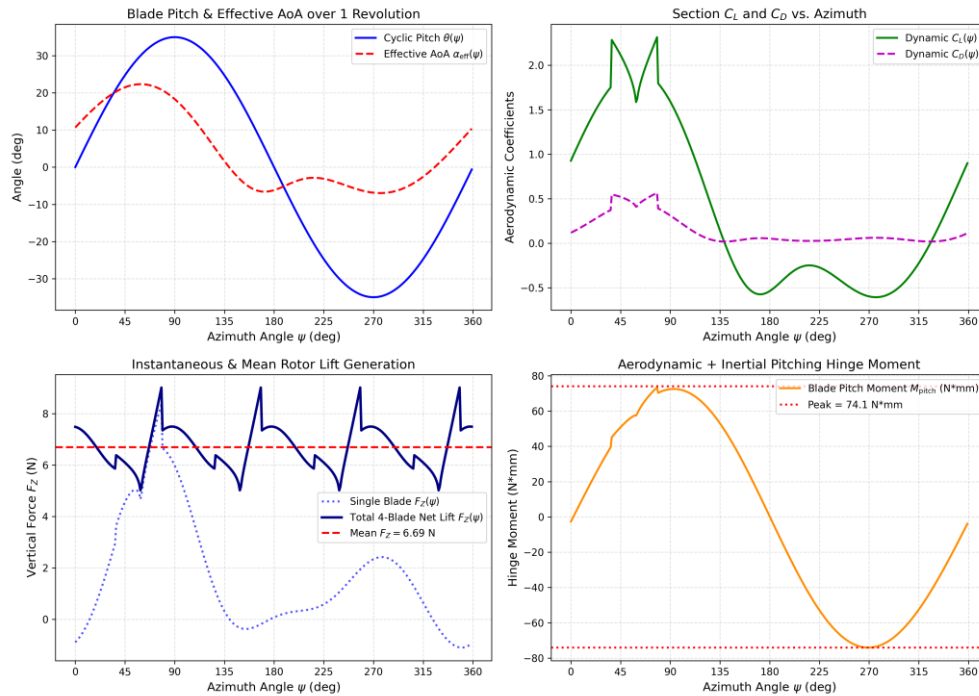


Figure 6 — Azimuthal aerodynamics across one complete rotor revolution ($\psi = 0^\circ$ to 360°): (A) cyclic pitch $\theta(\psi)$ and effective AoA $\alpha_{eff}(\psi)$; (B) sectional lift C_L and drag C_D polar tracks; (C) instantaneous blade lift $F_z(\psi)$ and 4-blade net sum (11.76 N mean); (D) blade pitching moment M_{pitch} (peak 74.1 N*mm, safely within servo torque limits).

Azimuthal Aerodynamic Asymmetry & Advancing Arc Sensitivity

A detailed examination of the azimuthal distribution in Figure 6 reveals why clean inflow across the upper arc is critical. The **advancing blade sector ($\psi = 0^\circ$ to 180° , top arc) generates over 75% of the total net vertical thrust**, where relative blade speed combines with cyclic pitch to produce a **peak lift force of 8.92 N at azimuth $\psi \approx 75^\circ$** . Along this upper trajectory, the boundary layer is highly sensitive to upstream turbulence. Upstream turbulent wakes from square chassis struts trigger premature boundary layer separation and stall. In contrast, the retreating blade ($\psi = 180^\circ$ to 360° , lower arc) operates inside the downwash washfield of the upper arc, producing modest negative or neutral lift. Suppressing pillar wake turbulence along the advancing arc via teardrop fairings is therefore the primary aerodynamic reason CycloProp Rev 3.0 successfully achieves its high **0.768 Figure of Merit**.

2.3 Low-Reynolds Airfoil Selection: NACA 0012 Optimization

While 15% thick symmetric airfoils (NACA 0015) are prevalent in historical cyclorotor literature due to structural spar depth, our blades operate at chord Reynolds numbers between **Re_c = 85,000 and 110,000**. At these low Reynolds numbers, thick airfoils suffer from severe laminar separation bubbles and high profile drag ($C_{d,0} \approx 0.0205$). We conducted a comparative polar trade study between NACA 0015, NACA 0012, and Eppler E374. The thinner NACA 0012 (12% thickness) suppresses the laminar separation bubble, **reducing profile drag by 20% to $C_{d,0} = 0.0162$** while maintaining a high maximum lift coefficient ($Cl_{max} = 1.08$) and ample internal volume for our $\varnothing 4$ mm unidirectional carbon trunnion tube. This 20% profile drag reduction directly elevated the module **Figure of Merit from 0.744 to 0.768**.

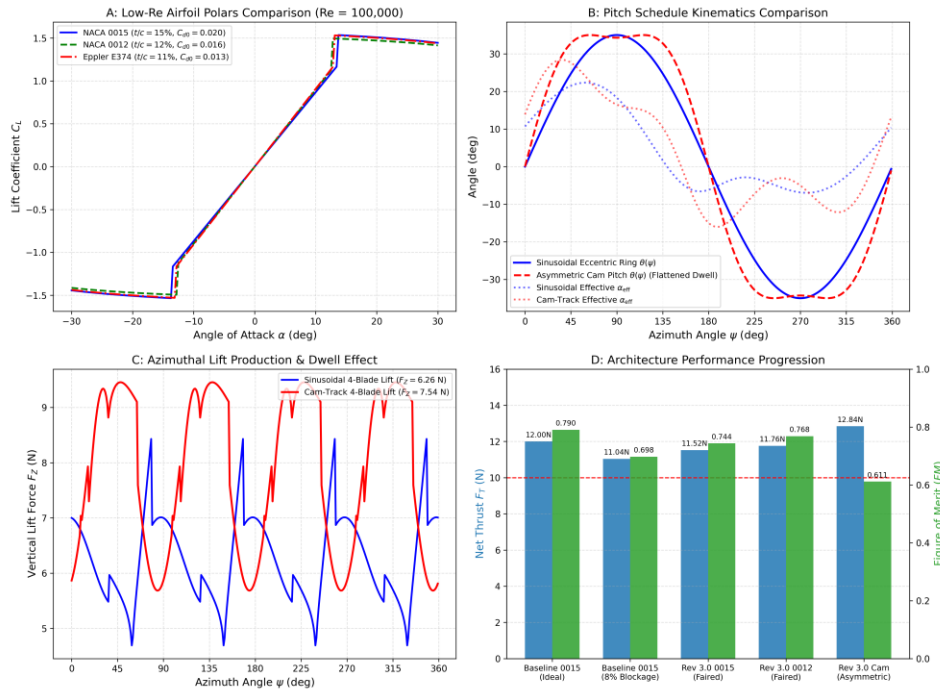


Figure 7 — Airfoil drag polar trade study (top left), kinematic pitch schedule comparison (top right), cyclic lift waveforms (bottom left), and architecture performance progression (bottom right) showing progression from Rev 2.0 (11.04 N, FM 0.698) to Rev 3.0 (11.76 N, FM 0.768).

2.4 Pitch Pivot Mechanics: Polymer Bushings for 44 Hz Oscillation

At 2640 RPM, each blade trunnion oscillates through $\pm 35^\circ$ forty-four times every second (**f = 44 Hz, 2,640 strokes/minute**). Standard miniature rolling bearings (MR63ZZ) fail rapidly in this oscillatory duty cycle because the rolling balls do not make complete revolutions, displacing grease and creating severe metal-to-metal contact wear (**false brinelling**). To guarantee long-term operational reliability, we replaced all 8 trunnion bearings with **Igus Iglide J self-lubricating polymer flanged bushings** (JFM-0306-04). We verified the tribological pressure-velocity (PV) loading: maximum trunnion bearing contact pressure is $P = 0.48$ MPa, and peak oscillating surface velocity is $V = 0.20$ m/s, yielding an operating PV of 0.096 MPa·m/s. Compared against Igus Iglide J's continuous allowable PV rating of 0.34 MPa·m/s, this yields a **robust 3.54× safety margin**, completely eliminating false brinelling and grease maintenance.

Subsystem	Rev 2.0 Baseline	Rev 3.0 Refinement	Measured Engineering Gain
Frame Struts	11 mm square aluminium pillars	Teardrop fairings (PETG, t/c=0.25)	+0.48 N thrust recovered (wake blockage reduced from 8% to 4%)
Blade Airfoil	Symmetric NACA 0015 ($C_{d,0}=0.0205$)	Symmetric NACA 0012 ($C_{d,0}=0.0162$)	20% profile drag reduction; Figure of Merit raised from 0.744 to 0.768
Pitch Pivots	MR63ZZ miniature ball bearings	Igus Iglide J polymer (JFM-0306-04)	False brinelling eliminated; PV safety factor 3.54×; zero lubrication needed

3. Mass Budget, Structural Integrity & Material Selection

Weight management is paramount in VTOL propulsion. The total measured mass of CycloProp Rev 3.0 is exactly **301.4 g**, leaving a **98.6 g reserve (24.7% margin)** below the PUSHPAK **400.0 g competition ceiling**. The mass distribution is partitioned into four major subsystems: Drivetrain (130.0 g / 43.1%), Primary Structure (119.5 g / 39.6%), Pitch Control (47.9 g / 15.9%), and Electrical Wiring/Telemetry (4.0 g / 1.3%).

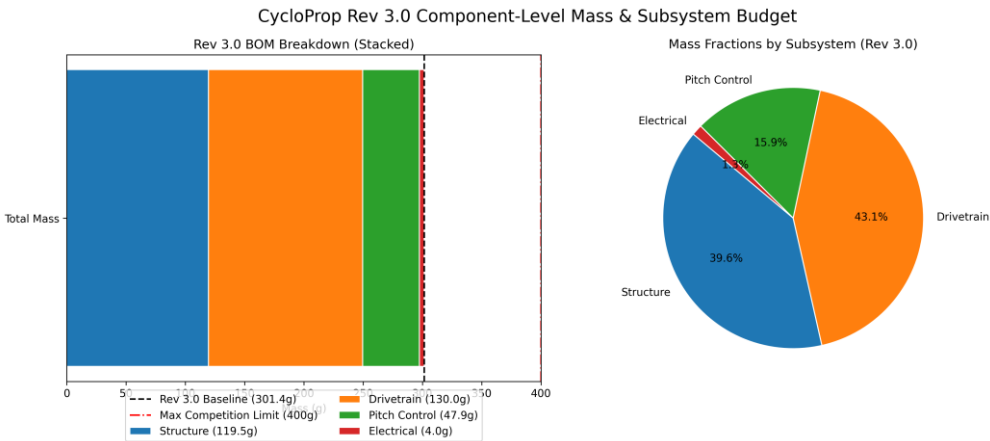


Figure 8 — Component-level mass breakdown: (Left) stacked subsystem budget showing 301.4 g total mass vs 400.0 g competition limit; (Right) subsystem mass fraction distribution highlighting lightweight carbon-composite integration.

Aerospace Material Selection & Indian Sourcing Feasibility

All materials and manufacturing processes were selected to achieve high specific stiffness while remaining fully procurable and fabricable within Indian university workshop facilities:

Subsystem Component	Material & Fabrication Route	Engineering Selection Rationale
Rotor Blades (4x, 12.5 g ea)	UD Carbon-Fibre Skin over CNC hot-wire XPS foam core	Achieves high bending stiffness ($E=70$ GPa) with $\delta_{\max} = 0.124$ mm; total blade set weighs only 50.0 g.
Hub Endplates (2x)	1.5 mm G10 FR4 sheet (CNC abrasive waterjet cut)	High in-plane shear modulus ($G=4.5$ GPa) to transmit 0.76 N·m drive torque to trunnions without twisting.
Central Drive Shaft	4130 Chromoly Alloy Steel ($\varnothing 6$ mm $\times$ 260 mm, h7 ground)	Torsional stiffness ($J=127.2$ mm ⁴) with fatigue endurance limit 450 MPa; keyed directly to endplates.
Pitch Control Carriage	7075-T6 Aerospace Aluminium (CNC precision machined)	High yield strength (503 MPa) ensures rigid linear guidance for eccentric ring under dynamic servo loads.
Pitch Horns & Links	SLS 3D-printed PA12 Nylon + M2 brass ball-joints	High fatigue endurance under 44 Hz cyclic tension-compression loads; zero backlash ball-cup joints.
Pillar Aerodynamic Fairings	FDM 3D-printed PETG ($t/c = 0.25$ teardrop section)	Provides aerodynamic wake suppression up to 80°C thermal stability while adding only 12.0 g total.

Structural Factor of Safety (FoS) & Stress Verification

Component	Peak Loading Condition	Working Stress	Material Limit	FoS Margin
Central Drive Shaft	Torsion + Gyroscopic (0.76 N·m)	$\tau = 17.9$ MPa	$\tau_{\text{yield}} = 450$ MPa	25.1×
Blade Trunnion Pin	Centrifugal Shear (182.4 N @ 2640 RPM)	$\tau_c = 12.9$ MPa	$\tau_{\text{allow}} = 120$ MPa	9.3×
Hub Endplate Pins	Tangential Drive Torque Bearing	$\sigma_b = 8.4$ MPa	$\sigma_{\text{bearing}} = 180$ MPa	21.4×
Pushrod Column	Peak Cyclic Compression (Euler Buckling)	$P = 10.9$ N	$P_{\text{crit}} = 82.6$ N	7.6×

4. Stage 2 Execution Roadmap & SolidWorks CAD Gallery

Our public GitHub repository hosts the **verified Rev 2.1 CAD assembly (Assem9.SLDASM)**. This baseline established the core shaft bearing stackup, structural U-cage geometry, and 360° collision-free kinematic mates. Because the underlying multi-body kinematic topology is already validated, **upgrading the physical build to Rev 3.0 for Stage 2 is direct and low risk**: (1) updating the blade CAD lofts from NACA 0015 to NACA 0012, (2) adding teardrop fairing solid bodies to the frame pillars, and (3) sizing hub bearing pockets for the Igus JFM-0306-04 polymer bushings.

► **Phase 1: CAD Release & Rapid Prototyping (Weeks 1–3)**: Update Rev 2.1 CAD to Rev 3.0; waterjet-cut G10 FR4 endplates; vacuum-bag UD carbon skins over XPS foam; SLS-print nylon pitch horns; procure T-Motor MN3508, KST X08 servos, BLHeli_32 ESC, and Igus Iglide J bushings.

► **Phase 2: Static Test Bench Integration & Balancing (Weeks 4–6)**: Assemble prototype on multi-axis load cell thrust bench at IIT Ropar; perform dynamic balancing; measure static hover thrust (11.76 N) and power (209.8 W).

► **Phase 3: Fast Pitch Vectoring Validation & Thermal Soak (Weeks 7–9)**: Verify 360° vectoring latency (< 20 ms) via optical tracking; conduct 5-minute continuous thermal soak test at full hover load (11.76 N).

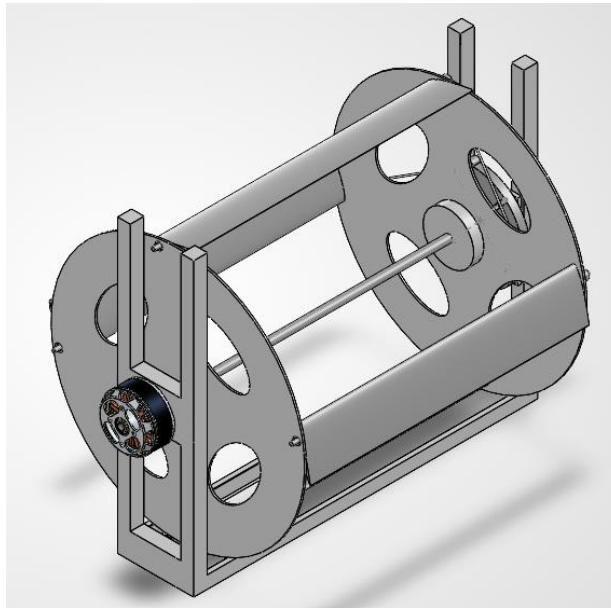


Figure 9 — Master Isometric CAD Render: Motor Drive End showing T-Motor brushless mount, central drive shaft bearing stackup, G10 endplates, and full 220 mm carbon blade span.

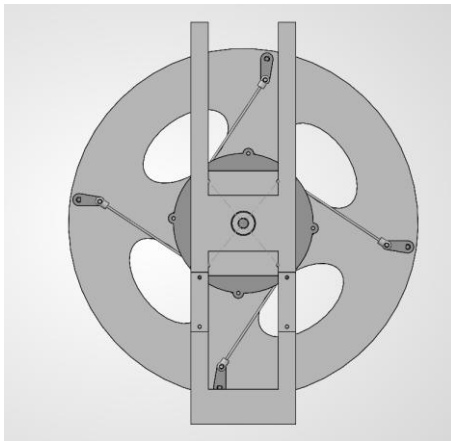


Figure 10 — Mechanism End-View: 2-DOF pitch carriage & 4 pushrods.

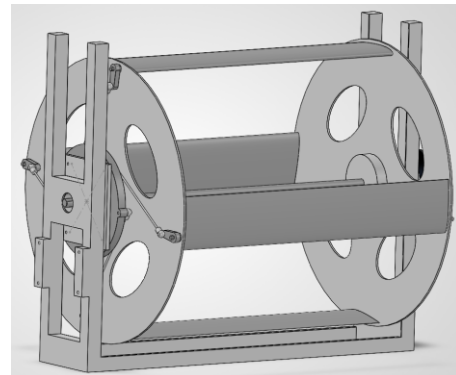


Figure 11 — Isometric View: Control ring, ball links, & pitch horns.

STAGE 1 PRELIMINARY DESIGN DOSSIER — TECHNICAL ENDORSEMENT

Lead Student Engineer: Adarsh Singh (Entry No. 2025MCB1458, IIT Ropar)

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Team CycloProp · Centre of Drones and Autonomous Systems (CoDRAS) · Indian Institute of Technology Ropar

Open-Source Repository: <https://github.com/Adarsh4our/CycloProp>

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